Hi

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Entanglement Dynamics in a Randomly-Kicked Ising model

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Kicked Ising model

The Hamiltonian for a 1D transverse field kicked Ising model is:

$$H(t) = \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x + \sum_{k=-\infty}^{\infty} \delta(k - \frac{t}{T}) \sum_{j=1}^{L} \sigma_j^z \qquad (1)$$

Floquet Operator

$$U = \exp(-iT \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x) \exp(-iT \sum_{j=1}^{L} \sigma_j^z) \tag{2}$$

Floquet Operator [2]

$$U = \exp(-i\frac{\pi}{4}\sum_{j=1}^{L}\sigma_j^x\sigma_{j+1}^x)\exp(-i\frac{\pi}{4}\sum_{j=1}^{L}\sigma_j^z) \tag{3}$$

Entanglement Entropy[2]

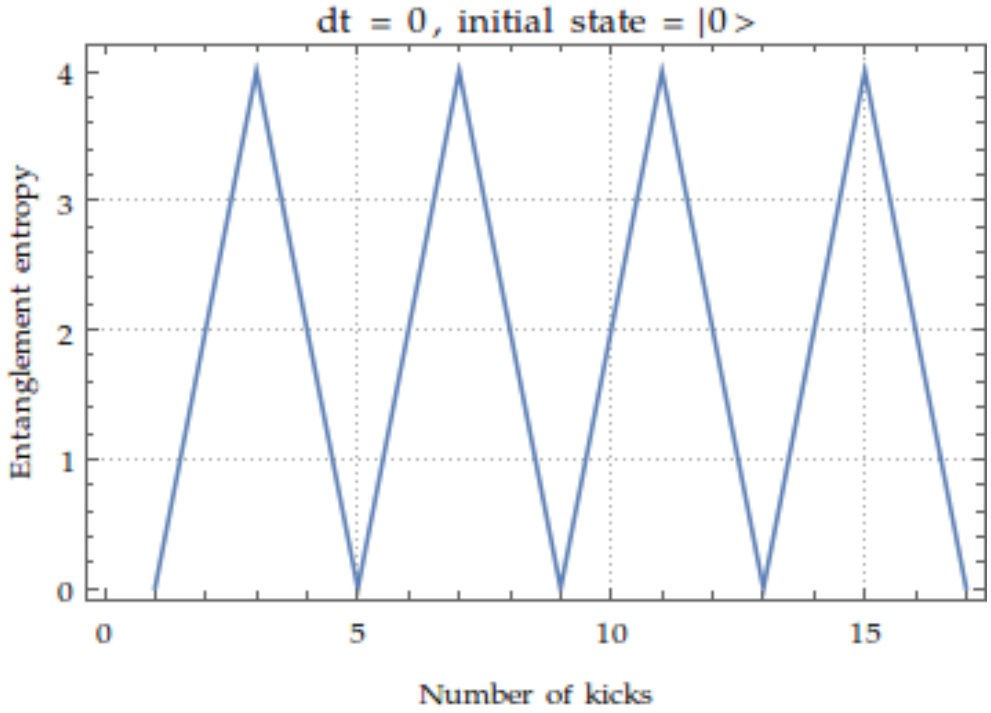


Figure: Starting with the initial state $|0\rangle$ in a 8-spin chain, entanglement entropy of the subsystem consisting of first 4 spins is being plotted with time for dt = 0

Jordan Wigner Transformations

Express the system in Fermion-momentum basis:

$$H_{kick} = 2\Sigma_k [d_k^\dagger d_k - d_{-k}^\dagger d_{-k}] \tag{4}$$

$$H_{after} = -2\Sigma_k (\cos k [d_k^\dagger d_k - d_{-k}^\dagger d_{-k}] - \sin k [d_{-k} d_k + d_k^\dagger d_{-k}^\dagger]) \tag{5}$$

Need for further simplification

Define a new space whose basis states are $|0\rangle$ and $d_k^\dagger d_{-k}^\dagger |0\rangle$.

$$H_{kick} = -2\Sigma_k \sigma_k^z \tag{6}$$

$$H_{after} = 2\Sigma_k (\cos k\sigma_k^z - \sin k\sigma_k^x) \tag{7}$$

Floquet Operator Now

$$U_{kick} = \exp(2iT\Sigma_k \sigma_k^z) \tag{8}$$

$$U_{after} = \exp(-2iT\Sigma_k(\cos k\sigma_k^z - \sin k\sigma_k^x)) \tag{9}$$

Very Simple Floquet operator!

When $T = \pi/4$,

$$U = -\prod_{\otimes k}(\cos k + i \sin k \sigma_k^y) \tag{10}$$

Random protocol[1]

$$U_n = U_{\text{after}}(U_{\text{kick}})^{\alpha_n}, \qquad \alpha_n \epsilon \{0, 1\}$$

Evolution on the Bloch sphere

Consider a single k-mode:

$$\begin{bmatrix} v_k(t) \\ u_k(t) \end{bmatrix} = e^{i\beta}\cos[\frac{\theta}{2}]|0\rangle + e^{i\alpha}\sin[\frac{\theta}{2}]|1\rangle \quad (11)$$

$$= e^{i\beta}(\cos[\frac{\theta}{2}]|0\rangle + e^{i(\alpha-\beta)}\sin[\frac{\theta}{2}]|1\rangle) \quad (12)$$

Evolution on the Bloch sphere, $T = \pi/4$

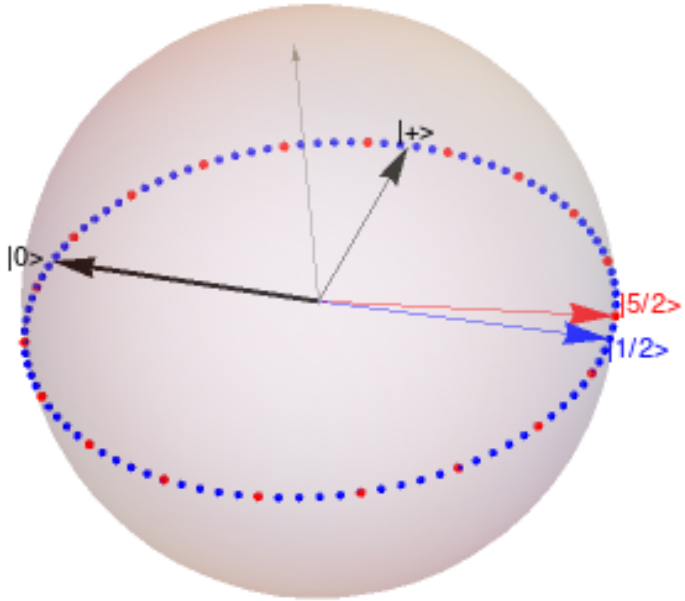


Figure: k-mode evolution, when $T = \pi/4$. The colored arrow shows the initial states for two modes $k = \pi/100$ and $k = 5\pi/100$

Evolution on the Bloch sphere, $T = \pi/4 + 0.01$

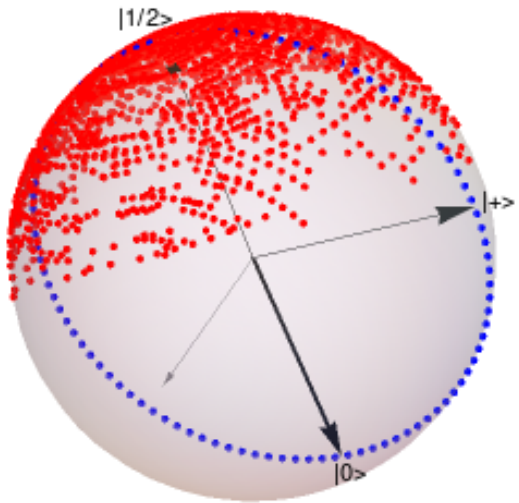


Figure: k-mode evolution, when $T = \pi/4 + 0.01$ shown in red for $k = \pi/100$ and the blue points correspond to $T = \pi/4$

Evolution on the Bloch sphere, $T = \pi/4 + 0.01$

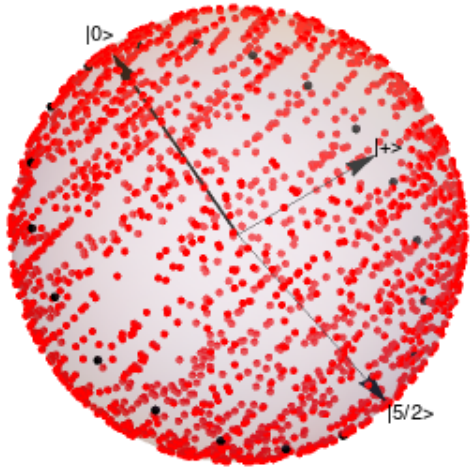


Figure: k-mode evolution, when $T = \pi/4 + 0.01$ shown in red for $k = 5\pi/100$ and the blue points correspond to $T = \pi/4$

Entanglement entropy

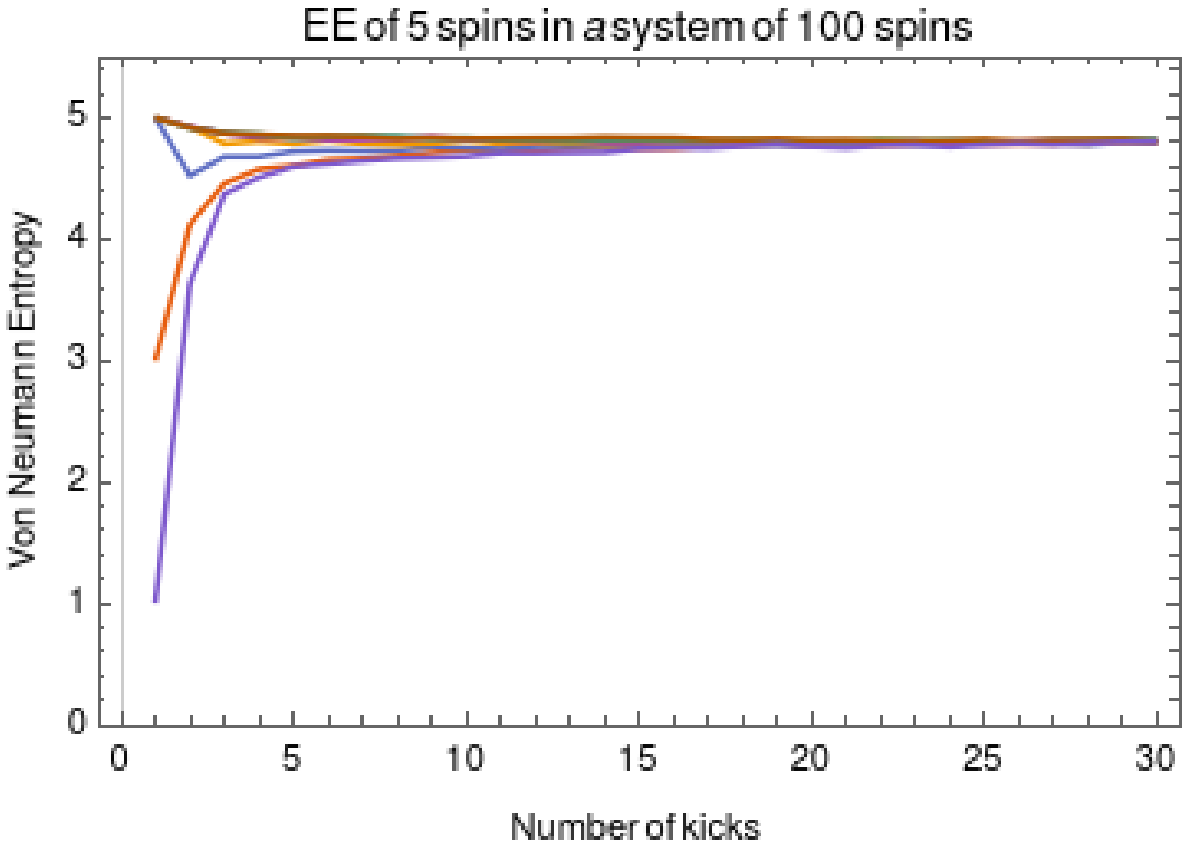
Von Neumann entropy:

$$\mathcal{S} = -\rho \log \rho \tag{13}$$

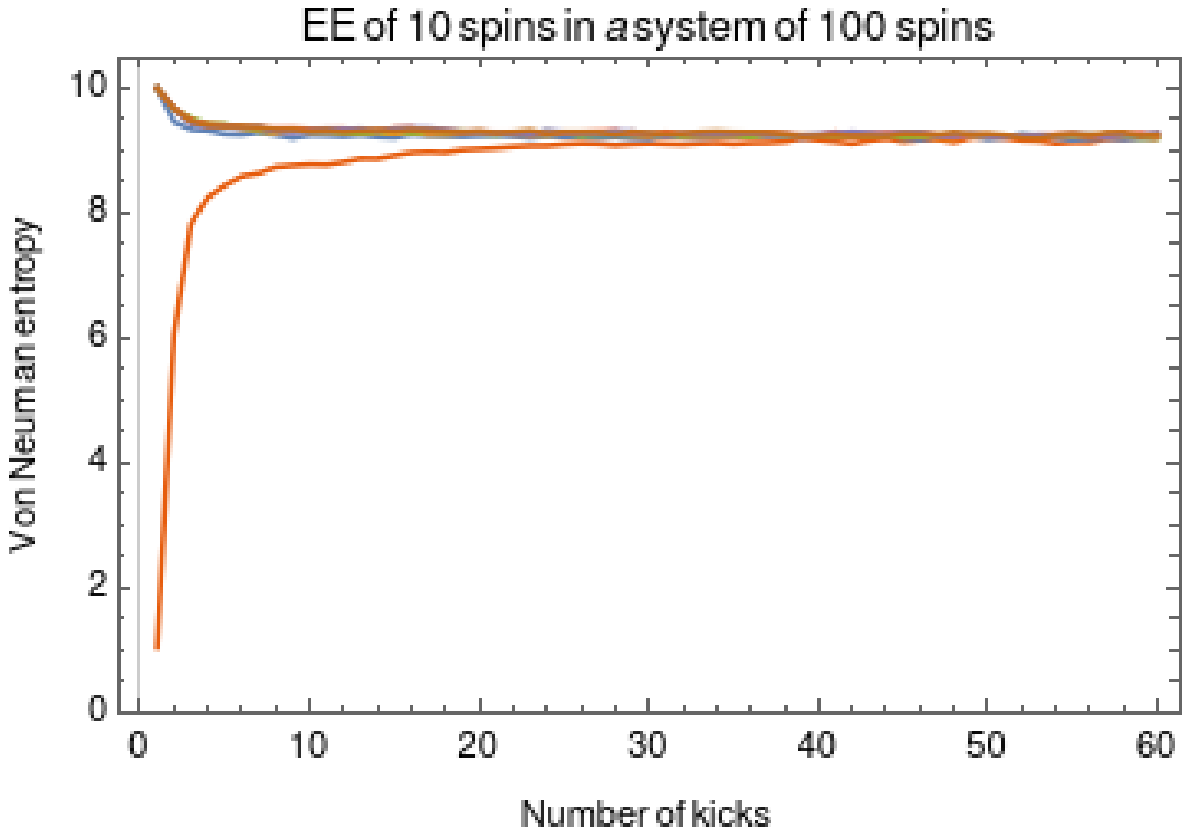
This is equivalent to the entropy of the eigen values of the correlation matrix given by[4][5]:

$$\mathcal{C} = \begin{bmatrix} I - C & F \\ F^\dagger & C \end{bmatrix} \text{ where, } \quad C_{mn} = \langle c_m^\dagger c_n \rangle \text{ and } F_{mn} = \langle c_m^\dagger c_n^\dagger \rangle \tag{14}$$

Entanglement entropy
Random case



Entanglement entropy
Random case



Remarks

- Special properties of $T = \pi/4$ has been previously explored for the periodic case [2]. Special behaviour in aperiodic dynamics could lead to better understanding of the system
- Thermalization of aperiodic systems as shown in [3] is not complete as we have seen an exceptional case.
- Further studies of Thue Morse/Fibonacci sequenced protocols can show interesting results

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Thank you!